Deepak Sathyan

Curriculum Vitae

arXiv | InspireHEP | Google Scholar | ORCID | Website

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Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

Experience

Postdoctoral Research Associate — College Station, TX
Mitchell Institute for Fundamental Physics and Astronomy — Sept 2024 – Aug 2027
Texas A&M University — 3 years

Education

University of Maryland — College Park, MD
Ph.D. in Physics — Aug 2018 – July 2024
Thesis: Unifying Searches for New Physics with Precision Measurements of the W Boson Mass
Advisor: Dr. Kaustubh Agashe

Boston University — Boston, MA
Bachelor of Arts in Physics, Summa Cum Laude with Honors — Aug 2014 – May 2018
GPA: 3.9
Minor: Mathematics
Thesis: Muon (g-2): Searching for the Muon Electric Dipole Moment
Advisor: Dr. Robert Carey

Research Interests

Beyond the Standard Model searches with Collider Physics: LHC and muon collider phenomenology, Randall-Sundrum and baryon-number-violation models

Searches for sub-GeV Dark Matter: dark matter-neutrino interactions, dark photon and dark sector production, cosmic ray-dark matter scattering

Neutrino Physics: supernova neutrino attenuation and detection, open-system observables, neutrinophilic mediators, neutrino NSI

Publications

Author order is alphabetical by last name, as per HEP convention.

6. New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis — July 2025
P.S.B. Dev, D. Kim, **D. Sathyan**, K. Sinha, Y. Zhang
arxiv.org/abs/2507.01000 (JHEP 10 (2026) 207)

5. **New laboratory constraints on neutrinophilic mediators** July 2024
   P.S.B. Dev, D. Kim, **D. Sathyan**, K. Sinha, Y. Zhang
   arxiv.org/abs/2407.12738 (Phys.Lett.B 868 (2025) 139765)

4. **'Unification' of BSM searches and SM measurements: the case of lepton+MET and m_W** Apr 2024
   K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, **D. Sathyan**
   arxiv.org/abs/2404.17574 (JHEP 02 (2025) 139)

3. **A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET** Oct 2023
   K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, **D. Sathyan**
   arxiv.org/abs/2310.13687 (Phys.Lett.B 855 (2024) 138774)

2. **Energy-peak based method to measure top quark mass via B-hadron decay lengths** Dec 2022
   K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, **D. Sathyan**
   arxiv.org/abs/2212.03929 (JHEP 06 (2023) 021)

1. **LHC Signals for KK Graviton from an Extended Warped Extra Dimension** Aug 2020
   K. Agashe, M. Ekhterachian, D. Kim, **D. Sathyan**
   arxiv.org/abs/2008.06480 (JHEP 11 (2020) 109)

Preprints

3. **Intensity-Frontier Signals of Warped Extra Dimensions** Aug 2026
   D. Kim, **D. Sathyan**, A. Verma
   arxiv.org/abs/2608.20482

2. **Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering** May 2026
   B. Dutta, D. Goswami, J. Kumar, M. Rai, **D. Sathyan**
   arxiv.org/abs/2605.08010

1. **A Baryon and Lepton Number Violation Model Testable at the LHC** Aug 2025
   A. Bhoonah, F. Burk, D. Liu, T. Ou, **D. Sathyan**
   arxiv.org/abs/2508.21064

Articles in Preparation

1. **LHC as a Beam Dump: Probing Light Dark Photons**
   B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, **D. Sathyan**
   (In preparation)

2. **New Probes of Dark Sectors at a Muon Collider**
   B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, **D. Sathyan**
   (In preparation)

3. **Exploring open systems approach to neutrino oscillations**
   A. Chandra Shekar, K. Kelly, **D. Sathyan**, L. Strigari, T. Zhou
   (In preparation)

4. **LHC Signals for KK Gravitons Producing 8-Particle Final States**
   K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, **D. Sathyan**
   (In preparation)

Snowmass2021 Contributions

5. **TF07 Snowmass Report: Theory of Collider Phenomena** — Oct 2022
   F. Maltoni et al.
   arxiv.org/abs/2210.02591

4. **Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021** — Sept 2022
   K. Agashe et al.
   arxiv.org/abs/2209.11267

3. **Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021** — Sept 2022
   T. Bose et al.
   arxiv.org/abs/2209.13128

2. **Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond** — Apr 2022
   K. Agashe, S. Airen, R. Franceschini, D. Kim, **D. Sathyan**
   arxiv.org/abs/2204.02928

1. **Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models** — Mar 2022
   K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, **D. Sathyan**
   arxiv.org/abs/2203.13305

Muon g-2 Articles

3. **The straw tracking detector for the Fermilab Muon g-2 Experiment** — Nov 2021
   B.T. King et al.
   arxiv.org/abs/2111.02076 (JINST 17 (2022) P02035)

2. **Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab** — Apr 2021
   Muon g-2 Collaboration
   arxiv.org/abs/2104.03240 (Phys.Rev.Accel.Beams 24 (2021) 044002)

1. **Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm** — Apr 2021
   Muon g-2 Collaboration
   arxiv.org/abs/2104.03281 (Phys.Rev.Lett. 126 (2021) 141801)

Ph.D. Thesis

1. Unifying Searches for New Physics with Precision Measurements of the W Boson Mass 2024
D. Sathyan
drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis)

Seminars

9. HEP Seminar at Northwestern University Sept 2026
Cosmic Ray–Dark Matter scattering explanation of the Galactic Center Excess

8. Fermilab Cosmic Physics Center Seminar Sept 2026
Cosmic Ray–Dark Matter scattering explanation of the Galactic Center Excess

7. CMS Exotica general meeting May 2024
Unification of Searches and Measurements: Probing BSM with the W boson mass measurement

6. Seminar at Texas A&M Dec 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

5. Seminar at UT Austin Dec 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

4. CMS SUSY general meeting Nov 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

3. HEP Seminar at University of Pittsburgh Nov 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

2. HEP Seminar at Northwestern University Oct 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

1. Theory Seminar at Washington University in St. Louis Oct 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

Conference Talks

14. Light Dark World 2026 at Carleton University [Invited] July 2026
Can the LHC be sensitive to the Light Dark World?

13. CETUP* Workshop [Invited] June 2026
Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering

12. Particle Physics on the Plains at the University of Kansas [Contrib.] Nov 2025
A Baryon and Lepton Number Violation Model Testable at the LHC

11. CETUP* Workshop [Invited] June 2025
Can the LHC be sensitive to light dark mediators?

10. Phenomenology Conference at the University of Pittsburgh [Contrib.] May 2025
Can the LHC be sensitive to light dark mediators?

9. Particle Physics on the Plains at the University of Kansas [Contrib.] Nov 2024
A comprehensive analysis of supernova neutrino-dark matter interactions

8. Texas TACOS @ UT Austin [Invited] Oct 2024
A comprehensive analysis of supernova neutrino-dark matter interactions

7. NuFact 2024 Conference at Argonne National Laboratory [Invited] Sept 2024
A comprehensive analysis of supernova neutrino-dark matter interactions

6. Mitchell Conference at Texas A&M University [Invited] May 2024
A comprehensive analysis of supernova neutrino-dark matter interactions

5. Particle Physics on the Plains at the University of Kansas [Contrib.] Oct 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

4. Phenomenology Conference at the University of Pittsburgh [Contrib.] May 2023
Probing Dark Matter-Neutrino Interactions via Supernova Neutrinos

3. Phenomenology Conference at the University of Pittsburgh [Contrib.] May 2022
Model-Independent Measurement of Top Quark Mass Using B-Hadron Decay Lengths (Part I)

2. April APS Meeting [Contrib.] Apr 2021
Signals of KK graviton from extended warped extra dimensions at the LHC (II)

1. Phenomenology Conference at the University of Pittsburgh [Contrib.] May 2020
Signals of KK graviton from extended warped extra dimensions at the LHC (II)

Teaching

Principles of Modern Physics Spring 2024
Teaching Assistant, University of Maryland

Electricity and Magnetism I Spring 2024
Teaching Assistant, University of Maryland

Mathematical Methods for Physics II Fall 2021, Fall 2022
Teaching Assistant, University of Maryland

Fundamentals of Physics I Summer 2019, Spring 2022
Teaching Assistant, University of Maryland

Introduction to Quantum Mechanics I (Graduate) Fall 2020
Teaching Assistant, University of Maryland

Fundamentals of Physics for Biologists II Spring 2019, Spring 2020
Teaching Assistant, University of Maryland

Fundamentals of Physics for Biologists Fall 2018, Fall 2019
Teaching Assistant, University of Maryland

Guest Lecture for Undergraduate E&M at University of Maryland Mar 2024
Lectured on vector potentials in E&M

Guest Lecture for Graduate E&M at Texas A&M University Oct 2025
Lectured on relativistic Larmor formula and charged particle motion in magnetic fields

Service

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2026
Conference Organizer — Texas A&M University, May 27-30, 2026

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2025
Conference Organizer — Texas A&M University, May 13-16, 2025

Mitchell Institute High Energy Phenomenology, Experiment, and Cosmology Seminar Series
Weekly Seminar Organizer - Texas A&M University, Fall 2024 - Present

Awards

Breakthrough Prize in Fundamental Physics Apr 2026
Awarded to the Muon g-2 Collaboration; recognized as a collaboration member

Ralph Myers & Friends of Physics Award Dec 2019
Awarded by the University of Maryland Department of Physics to outstanding Graduate Teaching Assistants

Ralph Myers & Friends of Physics Award May 2020
Awarded by the University of Maryland Department of Physics to outstanding Graduate Teaching Assistants

Alumni Award May 2018
Awarded by the Boston University Department of Physics

Schools Attended

GGI Lectures on the Theory of Fundamental Interactions Jan 2023

TASI 2024: The Frontiers of Particle Theory June 2024

Outreach

BU Student Mentorship Program PRISM 2016 - 2018
Mentored incoming freshman undergraduates in physics, providing guidance on courses, research opportunities, and resources

BU Outreach for Local High Schools Jan 2018
Demonstrated measurement of the charge-to-mass ratio of the electron for local high schoolers in physics

TAMU Physics and Engineering Festival 2025 - present
Presented demonstrations of fluid dynamics to the general public, including high school students and families

University of South Dakota talk: "My path through particle physics research so far" Dec 2024
Presented my path through particle physics research to students, showing a variety of interesting ideas to pursue

Skills

Programming:
Mathematica, Python, C++

Physics Software:
MadGraph5_aMC@NLO, Pythia6, Pythia8, Delphes, ROOT, GEANT4, FeynRules

References

Dr. Kaustubh Agashe
University of Maryland, College Park — kagashe@umd.edu

Dr. Bhaskar Dutta
Texas A&M University — dutta@tamu.edu

Dr. Bhupal Dev
Washington University in St. Louis — bdev@wustl.edu

Dr. Jason Kumar
University of Hawai'i — jkumar@hawaii.edu